

LAPTOP-BASED HMI For DC Motor Speed And Direction Control

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DC motors are usually controlled with switches or buttons for direction and a potentiometer for speed. While the

potentiometer remains effective for speed regulation, this system offers an alternative, enabling both speed and direction control of a DC motor (up to 36V) directly through a laptop. It integrates an Arduino Uno board for serial communication with a graphical user interface (GUI) developed using Processing. (Processing is a free, open source programming language and environment for creating interactive graphics, animations, and applications.) The GUI functions as a laptop-based human-machine interface (HMI), providing

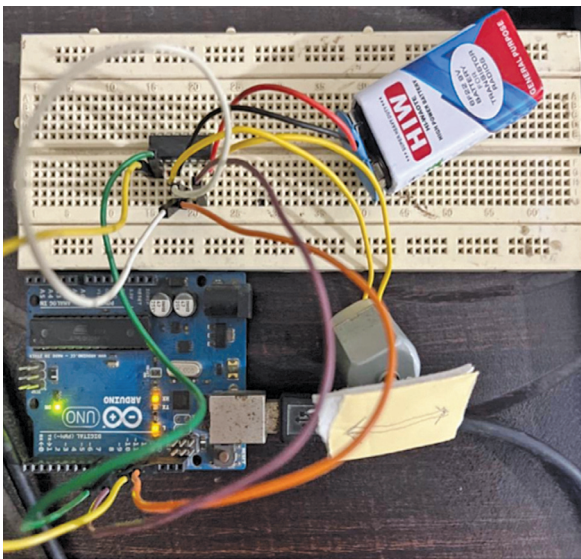


Fig. 1: Author's prototype tested on a breadboard

TABLE 1 BILL OF MATERIALS	
Components	Quantity
Arduino Uno board (MOD1)	1
L293D motor driver IC (IC1)	1
2-pin, 5.08mm pitch plug-in screw terminal block connector	1
DC motor (M) may use 5 to 36V DC	1
9V external power supply for motor	1
Laptop with USB interface	1
Connecting wires and a breadboard	Need based

smooth, screen-based motor control.

The components required to assemble the hardware are listed in the Bill of Materials Table 1. A breadboard-based author's prototype of the system is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the laptop-based HMI for DC motor speed and direction control. The system is built using an Arduino Uno board (MOD1), the L293D motor driver IC (IC1), a DC motor (M), and a few supporting components. A separate 9V power supply should be used for the motor instead of combining it with the Arduino's power source. The Arduino board requires a dedicated 9V supply. A USB cable is used for programming.

The system functions through serial

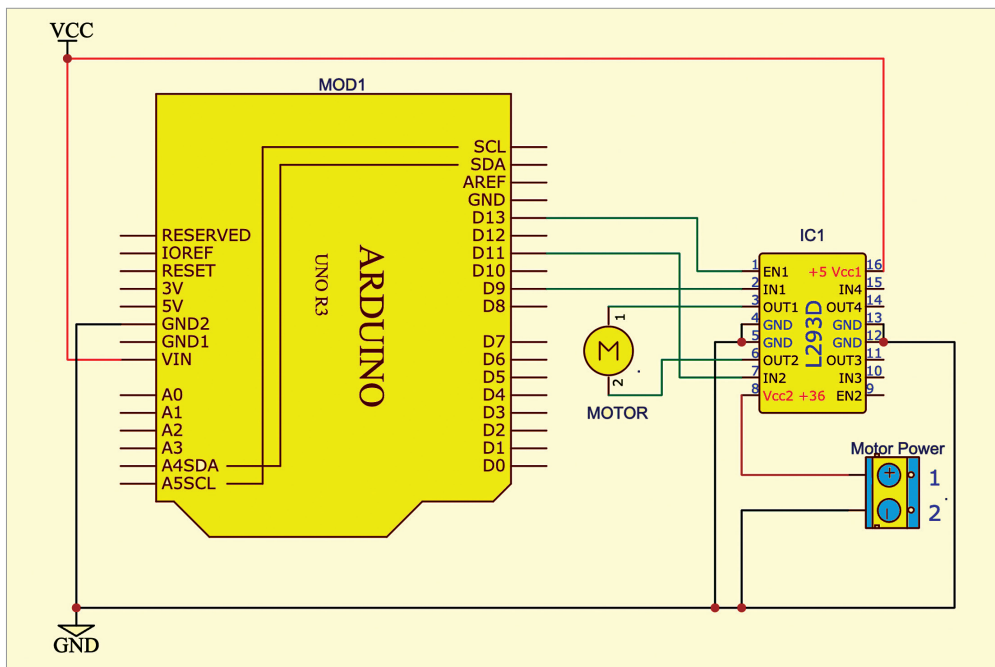


Fig. 2: Circuit diagram